

PROGRESS REPORT II

Purchase Order Data Extraction from Text or Images via Email using Machine Learning Techniques

ITC 4671 — Information Systems Development Research Project

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PRESENTATION OUTLINE

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04

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Process design, data pipeline & interface design

Introduction

Research Overview

- Organizations receive POs through email as unstructured text or image attachments
- Manual processing is time-consuming and error-prone
- Proposed system automates email classification, OCR, and entity extraction

Problem Statement

- Employees manually identify & extract PO details from hundreds of emails daily
- Variations in PO formats and low-quality images degrade accuracy
- Leads to delays, errors and high operational costs in procurement

Research Objectives

01 Investigate challenges in processing PO data from email communication

02 Examine ML & document processing techniques for email classification & PO extraction

03 Identify limitations of existing automated PO processing approaches

04 Design & implement an AI-based system integrating email classification, OCR & entity extraction

Data Collection Strategy

A combination of primary qualitative and primary quantitative data collection methods were used to gather requirements and datasets for the proposed system.

01 Physical Meetings

Face-to-face meetings with admin & technical staff to understand current PO email processing workflow

02 Virtual Meetings

Online meetings via Zoom with company supervisors & technical staff for requirements clarification

03 Observation

Direct observation of how employees identify and extract PO data from email content & attachments

04 PO Email & Image Dataset

Collection of sample PO emails and image attachments from the company for system development & testing

Method 1 & 2: Physical & Virtual Meetings

Method 1 — Physical Meetings		Method 2 — Virtual Meetings	
Purpose:	Understand current PO email workflow & identify operational challenges of manual data extraction	Purpose:	Clarify technical aspects of PO processing & discuss project requirements remotely
Participants:	Administrative staff (PO processing) Technical staff (email & documentation)	Participants:	Company supervisors Technical staff (email systems)
Data Collected:	PO email processing procedures, PO formats, issues in manual extraction, improvement suggestions	Data Collected:	Email communication workflow for POs Types of PO documents Technical integration considerations
Tool:	Face-to-face meetings at organization	Tool:	Zoom / online meeting platforms
Ethics:	Participants informed of academic-only use; no confidential info disclosed	Format:	Meeting notes & summarized discussions for requirement analysis

Method 3 & 4: Observation & Dataset Collection

Method 3 — Observation

Purpose:	Observe how employees manually process PO emails to document existing workflow
Target:	Staff responsible for reviewing & processing incoming PO emails
Observations:	How PO emails are identified, how PO details are extracted from body & attachments, steps for transferring data to internal systems
Data Type:	Qualitative observational data — existing manual workflow

Method 4 — PO Email & Image Dataset

Purpose:	Collect datasets for developing & testing the automated PO extraction system
Data Sources:	Sample PO emails from company PO image attachments via email Anonymized samples for testing
Data Types:	Email body text (unstructured) Image-based PO documents Scanned PO attachments (JPG, PNG, PDF)

Requirement Analysis

Business Requirements

- Automated system to process PO emails efficiently and reduce manual workload
- Handle both email body text and image-based PO attachments as input types
- Extracted data must be stored in a structured, exportable format (CSV/Excel)

Functional Requirements

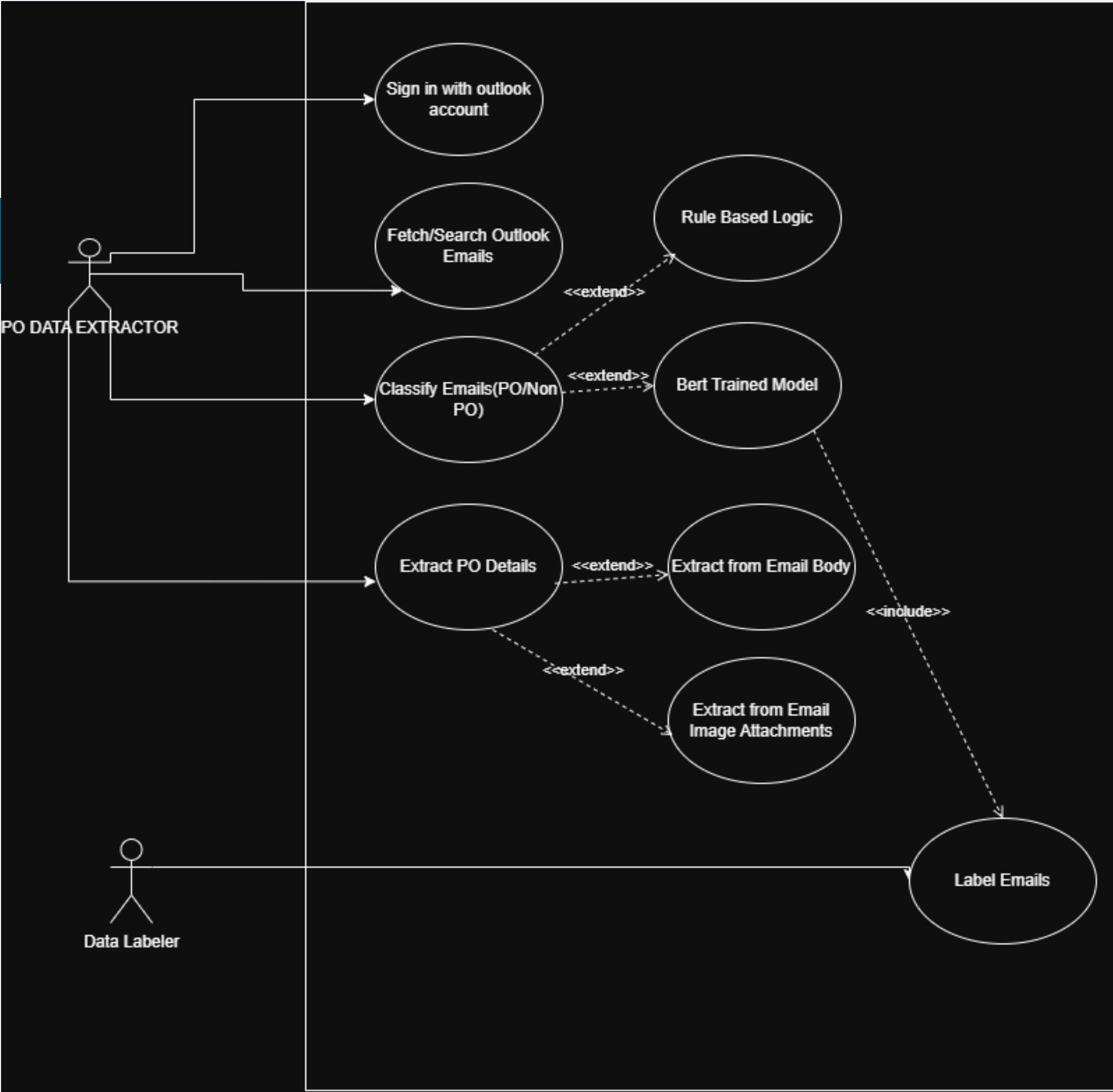
- Email Retrieval — secure authentication to email server
- Email Classification — ML model to identify PO emails
- Email Body Extraction — extract PO data from email text
- Attachment Processing — detect & process image attachments
- Image Preprocessing — enhance image quality for OCR
- OCR Text Extraction — extract text from PO images
- Entity Extraction — extract PO number, supplier, dates, quantities
- Structured Data Generation — organize data for export

Non-Functional Requirements

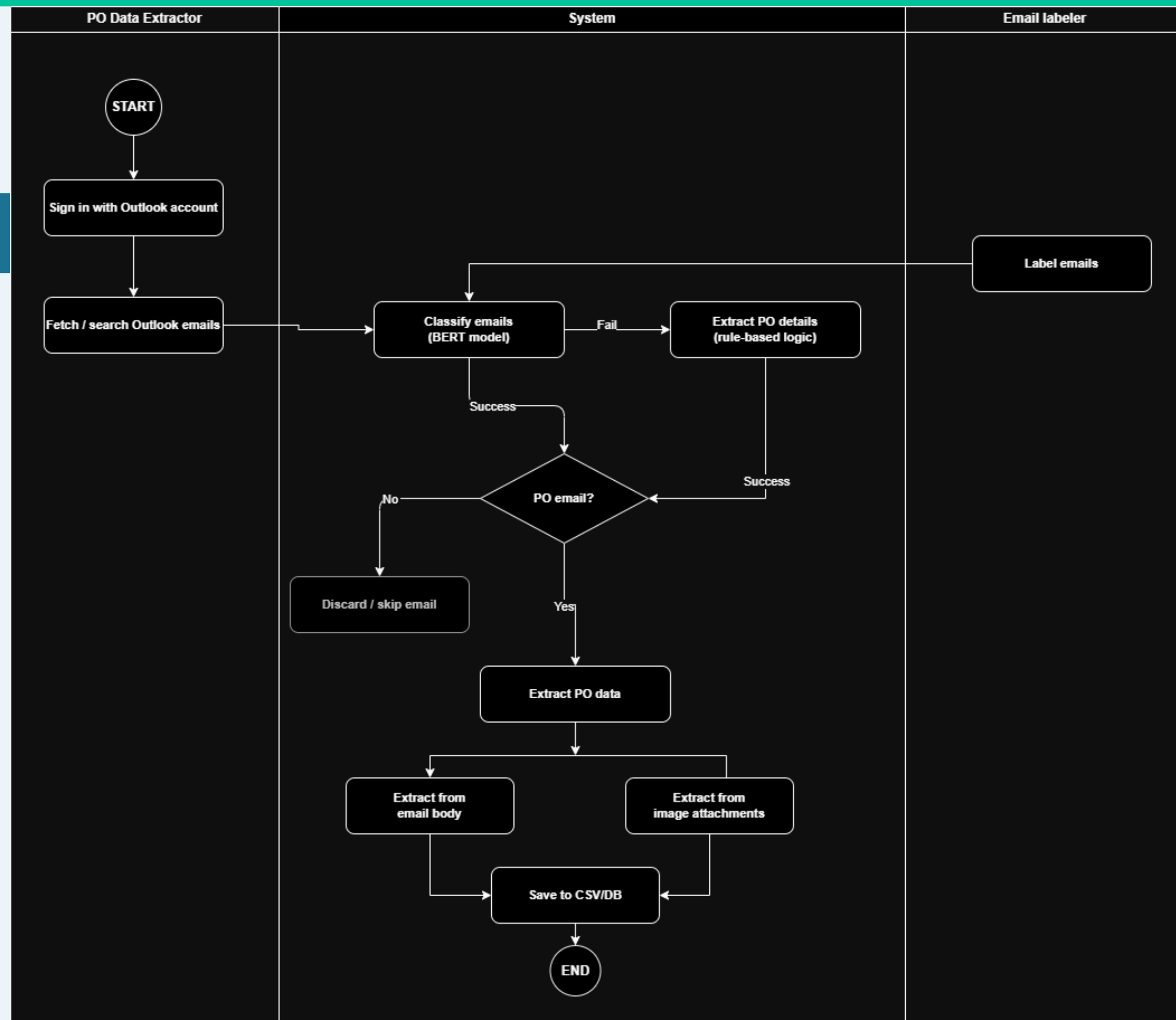
- Accuracy:** High precision in PO email classification & data extraction
- Performance:** Process emails within reasonable time for workflow efficiency
- Scalability:** Handle increasing PO email volumes as organization grows
- Security:** Secure handling of email data & sensitive information
- Reliability:** Stable operation across varied PO formats & structures
- Maintainability:** Allow future upgrades (e.g., LayoutLM for improved extraction)

System Design — Process Design

Use Cases



Activity Diagram — System Workflow

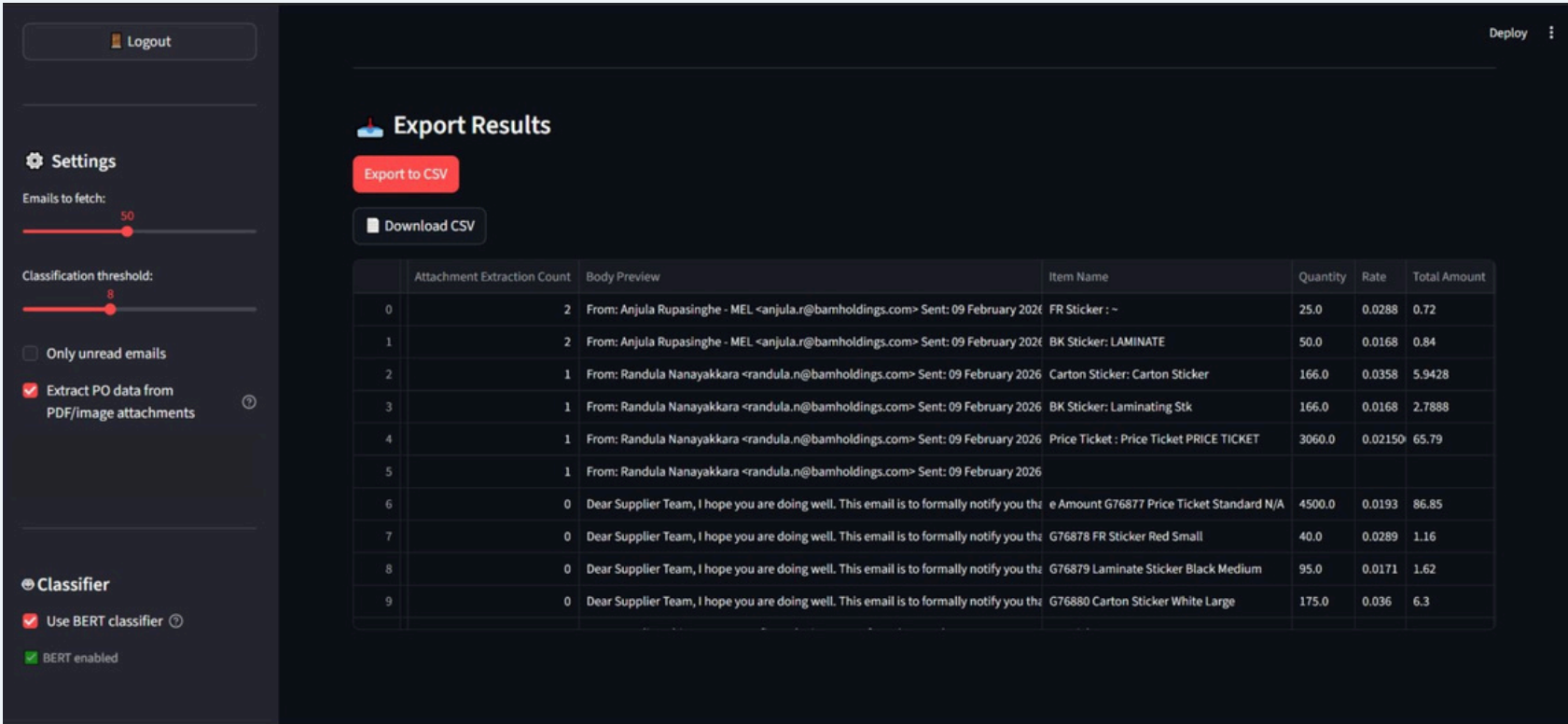
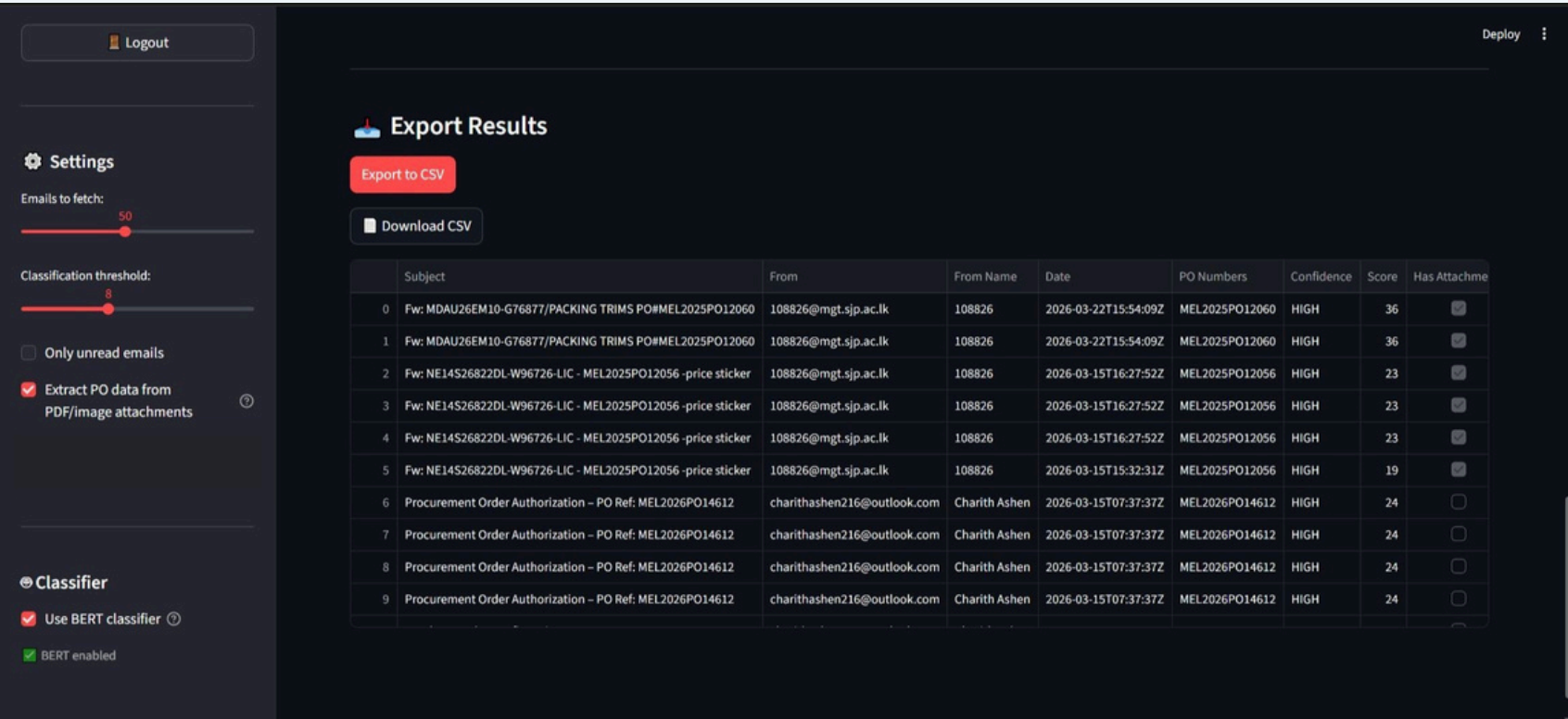
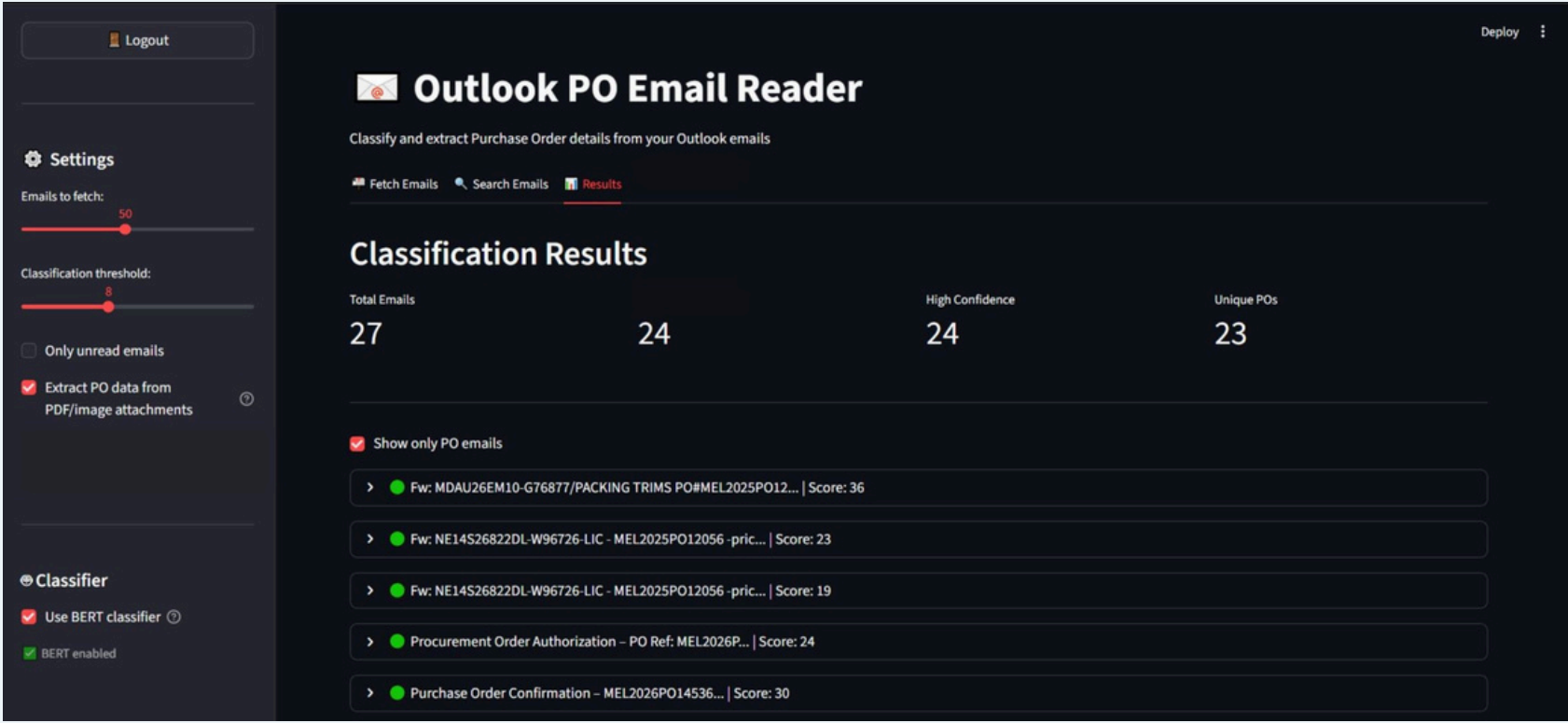
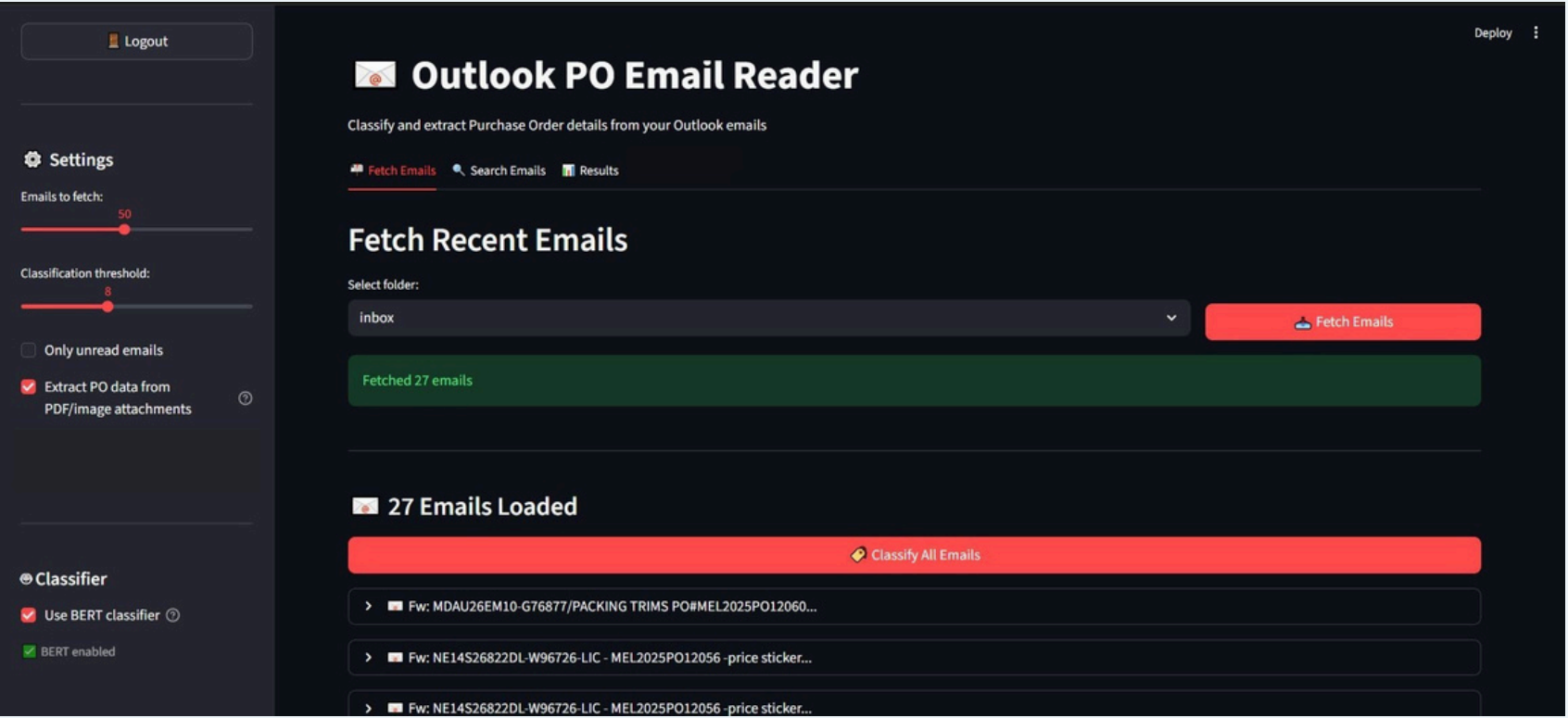


System Design — ML Pipeline & Interface Design

Machine Learning Data Pipeline



Interface Design Components



Thank You

Questions & Discussion

Project: Purchase Order Data Extraction from Email using ML Techniques